

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2458941.9307 +/- 0.0023 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1252 +/- 0.0025
Transit depth (Rp/Rs)^2	1.57 +/- 0.064 [%]
Orbital Inclination (inc)	82.98 +/- 0.59
Airmass coefficient 1 (a1)	0.9571 +/- 0.0046
Airmass coefficient 2 (a2)	0.0417 +/- 0.0053
Scatter in the residuals of the lightcurve fit is	0.48 %
Best Comparison Star	#2 - [151, 275]
Optimal Aperture	8.06
Optimal Annulus	15.95
Transit Duration (day)	0.0564 +/- 0.0029

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R fit vs published	0.1252 ± 0.0025 vs 0.10538 (deviation 1.14σ)
Duration fit vs published	0.0564 d vs 0.1167 d (deviation 2.99σ)
Mid-time O-C	2.7 min
Depth SNR	7.2
Residual scatter	0.48 %

Light curve

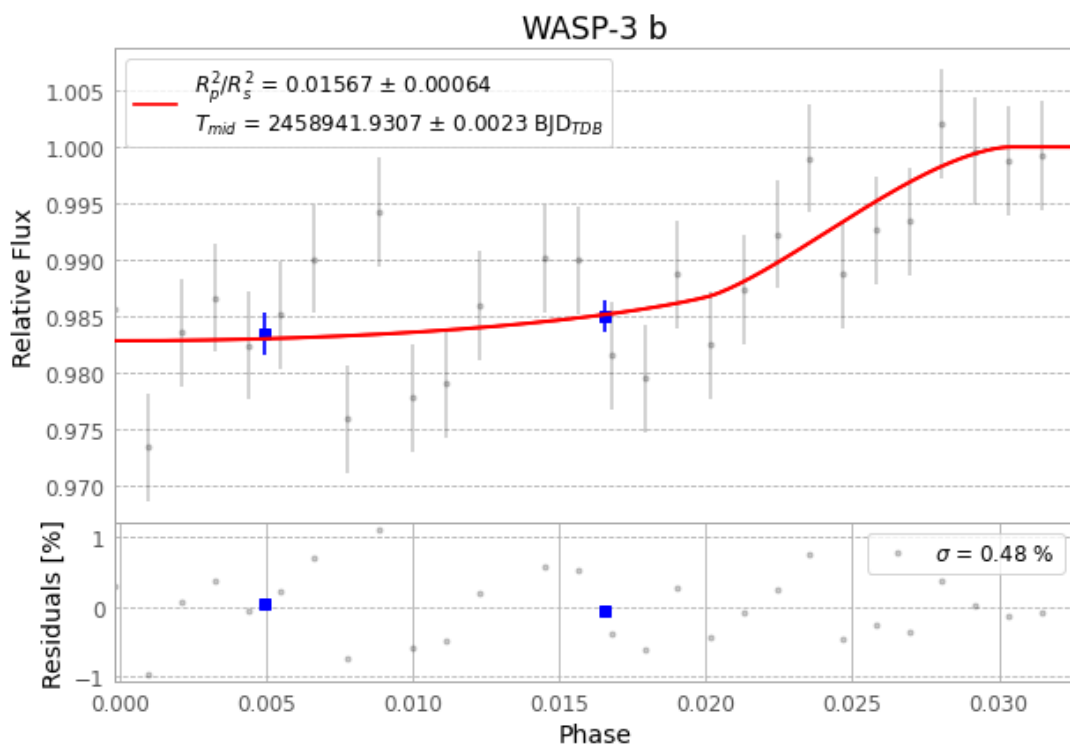


Figure 1 — FinalLightCurve_WASP-3 b_2020-04-02.png (SHA-256 6c5597951a8854e6...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [498, 228], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 e235b99f0aefb3a3...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

30 FITS frames (20 MB), WASP-3200402101812.FITS → WASP-3200402114514.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-3-200402.log (c9cb933381bb...), FinalLightCurve_WASP-3 b_2020-04-02.png (6c5597951a88...), FinalLightCurve_WASP-3 b_2020-04-02.csv (8a4092e85cef...)

Compute resources

Wall time 125.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 01:00Z.